

Neutron and X-ray scattering on Li-doped BPO₄

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Abstract

The structure of pure BPO₄ and Li-doped BPO₄ has been studied by neutron and X-ray scattering. Rietveld refinement of the neutron and X-ray spectra of the doped material did not show additional phases or a lithium superstructure. It appears that the interstitial Li⁺-ions as well as the charge compensating boron vacancies and substitutionally incorporated Li⁺-ions on boron sites are randomly distributed over the lattice resulting only in a slight change in peak intensities. The relevance of QENS for ceramic Li-ion batteries is discussed. © 1999 Elsevier Science B.V. All rights reserved.

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1. Introduction

The use of nano-structured Li-doped BPO₄ as a solid electrolyte for ceramic Li-ion batteries has been demonstrated by Kelder et al. [1]. The ionic conductivity depends on the Li-doping level and a maximum was found in the conductivity isotherm around 5 mol% lithium. Crystallographic studies of undoped (pure) BPO₄ stem from the late 30s [2] and revealed that the crystalline ceramic adopts the high-cristobalite structure. In the tetragonal unit cell all the B³⁺- and P⁵⁺-ions are tetrahedrally coordinated by O²⁻-ions, each O²⁻-ion being

shared by two tetrahedra (Fig. 1). The structure of BPO₄ has been described before by a tetragonal structure with the space group I-4 and lattice constants $a = b = 4.3425$ Å and $c = 6.6415$ Å [2,3].

The defect structure of Li-doped BPO₄ has recently been reported by Jak et al. [4] and is given by the defect models:

$$\begin{aligned} \text{(a)} \quad & V_B''' + 3Li_i^+ \\ \text{(b)} \quad & Li_B'' + 2Li_i^+ \end{aligned} \quad (1)$$

Here, the Kröger–Vink defect notation is used [4]. In (a) a boron vacancy (net charge 3 −) is surrounded on average by three interstitial Li⁺-ions. In (b) one Li⁺-ion is virtually fixed to a boron vacancy (net charge 2 −) and surrounded on average by two interstitial Li⁺-ions. In these two defect models the ionic conductivity arises completely

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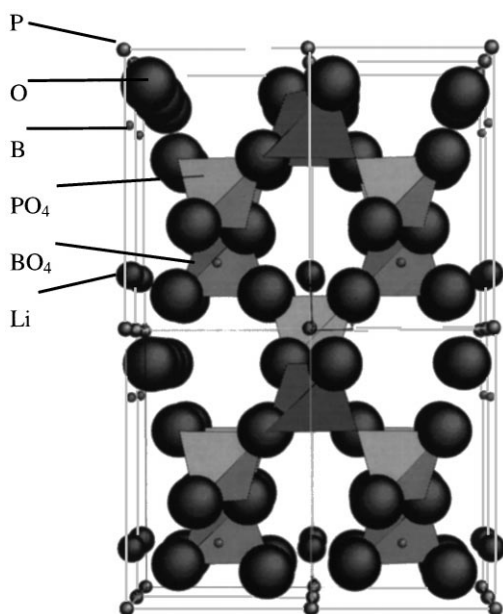


Fig. 1. High-cristabolite structure of Li-doped BPO_4 .

from the motion of the interstitial Li^+ -ions [4]. In practice one expects both processes (a) and (b) to be present in a dynamic equilibrium, i.e.



The ratio of occurrence of (a) and (b) is unclear at present.

In this paper the structures of pure BPO_4 and 10 mol% Li-doped BPO_4 were investigated as well as the locations of the interstitial Li^+ -ions, the boron vacancies and the Li^+ -ions on boron sites. We do so by means of neutron diffraction (ND) and X-ray diffraction (XRD). Both techniques are complementary, with ND more sensitive to the Li^+ -ions.

2. Experimental

The synthesis of BPO_4 with and without Li-doping is described in Ref. [5]. Equimolar amounts of boric acid (H_3BO_3) and phosphorus pentoxide (P_2O_5) were mixed in water in order to obtain a homogeneous white paste. For the synthesis of

Li-doped BPO_4 , 10 mol% of $\text{LiOH} \cdot \text{H}_2\text{O}$ was added to the precursors. The BPO_4 and 10 mol% Li-doped BPO_4 -samples were dried under vacuum for several hours.

Neutron diffraction (ND) measurements on the BPO_4 -powder samples were performed at room temperature on POLARIS, a time-of-flight powder diffractometer at the pulsed neutron source ISIS (England). A rectangular titanium–zirconium sample container with a width of 20 mm, a height of 50 mm, and a thickness of 2 mm was used. Detector banks were situated on both sides of the neutron beam. The flat sample container was placed in the beam such that the surface made an angle of 45° with the incoming neutron beam. Due to this configuration the absorption is different for both detector banks used.

The data have been corrected for absorption in such a way that after the correction the diffraction patterns, measured on different sides of the neutron beam, matched each other.

The X-ray powder diffraction (XRD) spectra were recorded using a Siemens D 5000 diffractometer with a Cu K_α -source equipped with a Göbel mirror. A step size of 0.01° and 10 s counting time per step were used.

The ND and XRD diffraction spectra were analyzed by means of the Rietveld refinement method, using the computer program general structure analysis system (GSAS) [6].

3. Results and discussion

The neutron spectrum of pure BPO_4 , measured at 90° scattering angle, is given in Fig. 2 as function of the d-spacing. Refining the data with the space group given in Ref. [3] results in an excellent fit. The lattice parameters $a = b$ and c are given in Table 1.

The ND-spectrum of the 10 mol% Li-doped BPO_4 reveals only small differences in intensities compared with the pure BPO_4 -sample. The absence of additional peaks in the spectrum indicates that no second phases are present. The spectrum of the Li-doped BPO_4 -sample is analyzed using the structure of pure BPO_4 , resulting in a good fit (Fig. 3). Only small deviations of the lattice

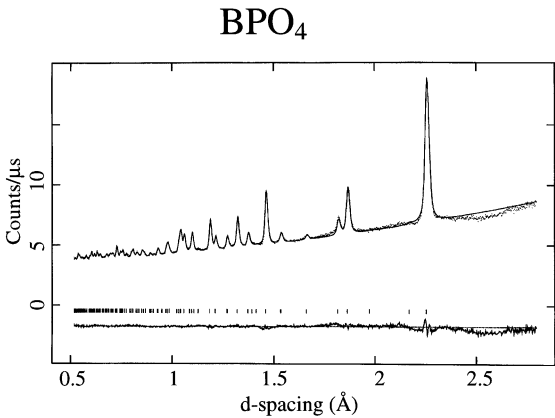


Fig. 2. Neutron diffraction spectrum of undoped BPO₄ (dots) with GSAS fit (solid line). The marks give the allowed peak positions.

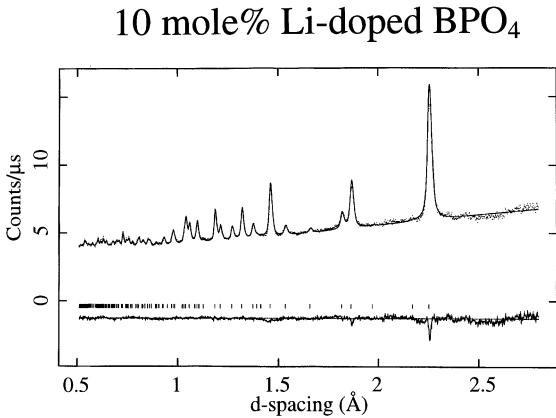


Fig. 3. Neutron diffraction spectrum of 10 mol% Li-doped BPO₄ (dots) with GSAS fit (solid line). The marks give the allowed peak positions.

Table 1
Lattice parameters of BPO₄ and 10 mol% Li-doped BPO₄ determined with ND and XRD

	<i>a</i> -axis (Å)	<i>c</i> -axis (Å)	Cell volume (Å ³)
JCPDS (0%Li) [3]	4.3425	6.6415	125.24
ND (0% Li)	4.3412	6.6472	125.28
XRD (0% Li)	4.3413	6.6479	125.28
ND (10% Li)	4.3426	6.6363	125.15
RD (10% Li)	4.3493	6.6487	125.77

constants compared with the undoped material are observed (cf. Table 1).

The XRD-spectra of pure BPO₄ show very good agreement with the reported data [3]. All the reported peaks are visible. Also no additional peaks are observed in doped BPO₄. Comparison of the XRD-spectra of the undoped and doped samples shows very slight differences in peak positions, due to small changes in the lattice parameters, and significant differences in peak intensities, or more precisely, the peak areas. These differences are caused by the loss of boron ions when lithium ions are incorporated in the structure of Li-doped BPO₄ [4]. In Table 1 the *a*- and *c*-axis of the all XRD- and ND-measurements are listed for the doped and undoped samples. As can be seen in Table 1, an excellent agreement between ND and XRD on undoped BPO₄ is found. On the other hand, the

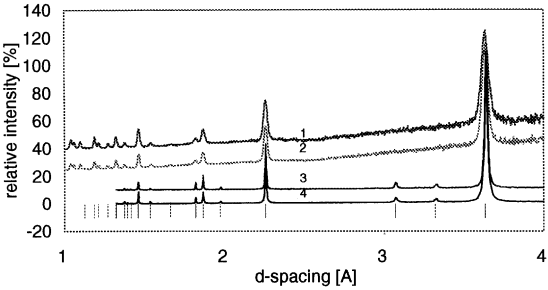


Fig. 4. Comparison of the ND- and XRD-spectra of undoped and Li-doped BPO₄. 1 = ND of 10 mol% doped BPO₄, 2 = ND of undoped BPO₄, 3 = XRD of 10 mol% doped BPO₄, 4 = ND of undoped BPO₄. The vertical markers are the literature data [3].

difference between ND and XRD on lithium doped BPO₄ is significant. An explanation for this has not yet been found. A different sample preparation method or different conditions during the experiments might cause such differences.

In Fig. 4 all the XRD- and ND-spectra are collected. Three observations can be made:

1. Two peaks are missing in the ND-spectra, namely the 110 (*d* = 3.0699 Å) and the 002 (*d* = 3.32107 Å) reflections.
2. There are no additional peaks in the ND-spectra compared with the XRD-spectra.
3. All remaining peaks of the four spectra show good matching in position.

The reflections missing in the ND-spectra can be explained by the different scattering lengths of the elements compared with XRD-measurements. Using in the computer program GSAS the scattering lengths of B and P as free varying parameters one finds destructive interference for the 110 and 002 reflections when the scattering lengths of B and P are about equal (as is the case for ND). These reflections are present when the scattering length of B is much smaller than the one of P (as is the case for XRD).

Absence of additional peaks compared with pure BPO_4 shows the absence of second phase and a superstructure involving lithium ions or boron vacancies. This result indicates that the interstitial Li^+ -ions are randomly distributed over the available lattice sites. The charge compensating boron vacancies or the substitutionally incorporated Li^+ -ions on boron sites are also randomly distributed. The only contribution of the charge compensating defects to the spectra is found in the intensity differences.

4. Conclusions

Neutron and X-ray powder diffraction measurements of undoped BPO_4 show good agreement with each other. The measurements on 10 mol% Li-doped BPO_4 show a slight change in lattice parameters and no second phases. The interstitial Li^+ -ions, the charge compensating boron vacancies and the substitutionally incorporated Li^+ -ions on boron sites are randomly distributed resulting in a slight change in peak intensities.

In the present experiment we could not determine the relative weight of the defect processes (a) and (b) given in Eq. (1). To do so in ND one first needs the boron isotope B^{11} . This isotope absorbs far less than natural boron, which makes the absorption smaller, leading indirectly to more precise results. Furthermore, the randomly distributed Li-ions cause diffuse scattering, from which their spatial distribution can be deduced. The accuracy of

the present measurements appears to be too low to observe diffuse scattering clearly. Finally one needs quasi elastic neutron scattering (QENS). With this technique one determines the microscopic motion of the lithium ions. A distinction can be made between jump diffusion between discrete sites and continuous diffusion. In the case of jump diffusion one determines the jump rates and the average distance between the discrete sites. For continuous diffusion one obtains the diffusion coefficient. Thus with QENS one can distinguish between the Li^+ -ions which are fixed to the boron vacancies and the interstitial moving ions. Hence one obtains the ratio of occurrence of the two processes (a) and (b) given in Eq. (1). Such QENS experiments are in preparation.

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